Diffraction

Diffraction: The bending of light round a sharp edge or a narrow opening or a very small obstacle is known as diffraction. By sharp narrow or small we mean that these sizes should be comparable to the wave length of light.

Depending on the position of the source and the screen with respect to the obstacle diffraction phenomenon can be classified into two.

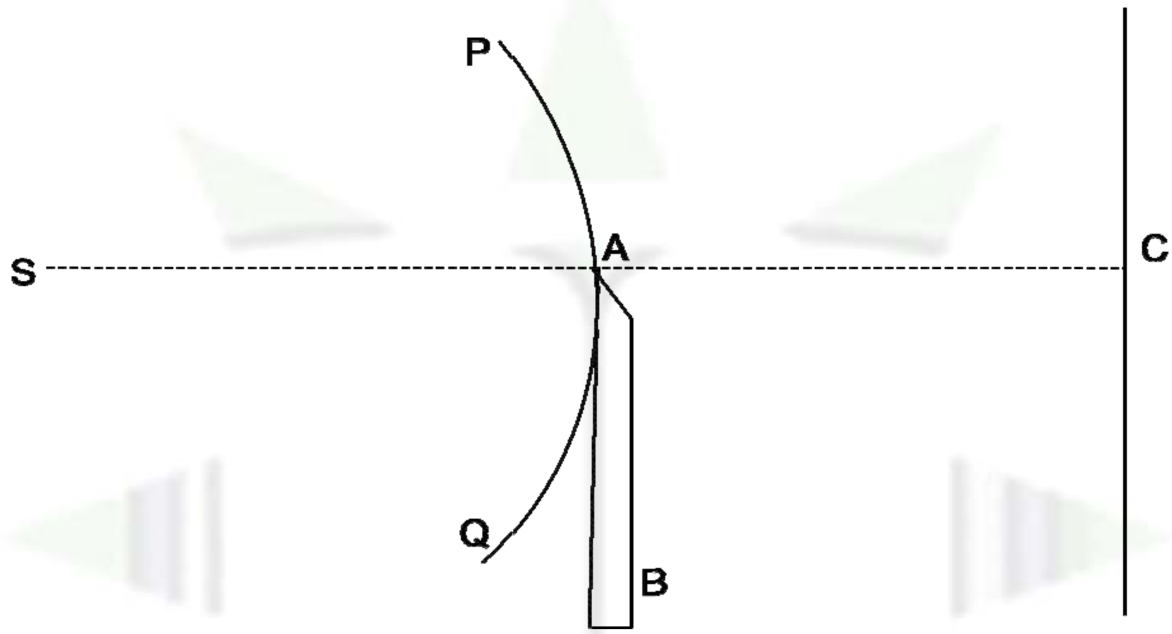
(1) Fresnel's Diffraction

(2) Fraunhoffer Diffraction

(1) Fresnel's diffraction: When both the source and the screen are at finite distance from the obstacle i.e. the incident wave front is either spherical or cylindrical but not a plane wave front the class of diffraction pattern is known as Fresnel's diffraction.

(2) Frounhoffer Diffraction: When both the source and the screen are at infinite distance from the obstacle i.e. the incident wave front is a plane wave front the class of diffraction pattern obtained is known as Frounhoffer diffraction.

Fresnel's diffraction at a straight edge: Let us consider a straight edge like edge of razor blade mounted vertically on a stand in an optical bench. A rectangular slit illuminated with sodium light is kept behind the straight edge, an eye piece mounted on a stand in front of straight edge acts as the prism.



AB: The section of the straight edge by the plane of the paper.

S: The section of the rectangular slit by the plane of the paper. The slit, straight edge and the screen all are perpendicular to the plane of the paper.

PQ is the section of the cylindrical wave front, incident on the straight edge by the plane of the paper. Joining the source S with the edge A which cuts the screen at C, line SAC defines the boundary of the geometrical shadow.

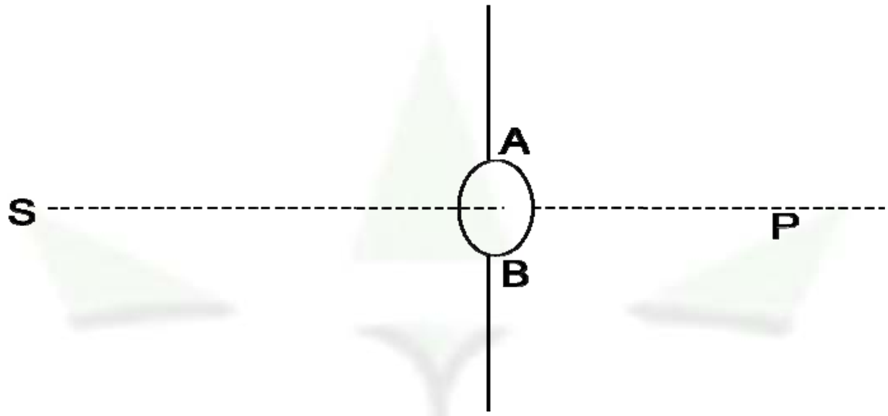
The following observations are made on the screen:

(1) Within the geometrical shadow i.e. below the line SAC in the figure instead of complete darkness some light intensity can be found. However the bright intensity decreases very rapidly as we move more and more into the shadow and vanishes within a very short distance.

(2) Outside the geometrical shadow i.e. above the line SAC in the figure instead of uniform illumination alternative bright and dark fringes can be seen. These fringes are not equispaced like interference fringes and the fringes come closer and closer as we move more and more away from

the shadow and finally merge to give general illumination. These fringes are known as diffraction fringes.

Fresnel's diffraction at a circular aperture:



A card board is placed with its plane vertical. A pin hole is punched on the board. A point source of light S is kept on the axis of the pinhole. P is any point on the axis on the opposite side of the pin hole.

The intensity at different points on the axis are found to be different and it changes from a maximum to a minimum and then to a maximum and so on. The intensity at the point P may be maximum or minimum that depends on the size of the aperture. Keeping the position of the point fixed if we change the diameter of the aperture the intensity at P also changes.